Sheriff case SHF-20261024-46b7da

Published-main validation snapshot, 24 October 2026. The authoritative machine-readable verdicts are in `findings.yaml`; the full narrative and limitations are in `REPORT.md`. This case changes no research status or experiment record.

Exact source scope

repository	published commit
crypto-autoresearcher	`4647b1d072ffaebbbb85127bdc60343bbd372ec4`
crypto	`172f84c7b0143ff5b4a26d6d21ffed3bf0f0cf3b`
cryptanalysis	`9d320ae7df3b8aa15d4497d6895f63644daa92d5`

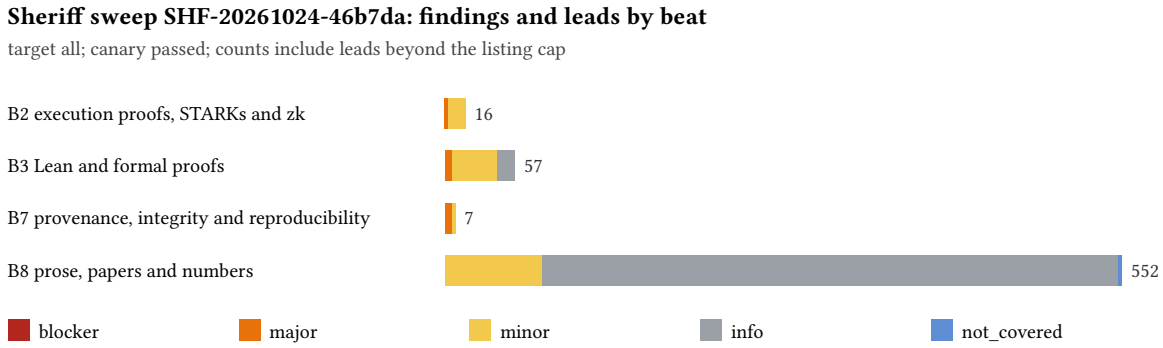

Figure 1: Deterministic deep-sweep signals. A scanner lead or a checker timeout is not a research verdict; see `findings.yaml` for scoped adjudication.

Scoped verdicts

ID	verdict	established scope
C1	scoped	84 / 84 pinned target declarations machine-verified; semantic fidelity pending
C2	scoped	conditional BINSTD arithmetic ceiling, with satisfiable premises
C3	uncovered	archived 40 / 40 EC execution not independently replayed
C4	scoped	binary polynomial product identity only
C5	scoped	3.2391 charged cold wall-time ratio, one host, two rank seeds
C6	scoped	approximately 94-fold cold gap to matched rho on this host
C7	scoped	tested degree-one STARK examples; general soundness open
C8	scoped	14 receipt-source hashes; selected Lean builds only
C9	broken	sheriff ecbench checker omits gzip expansion for archived sessions
C10	broken	two configured cryptanalysis Lean gate scripts are absent
C11	broken	formal checker timeout left an untracked Lean audit file
C12	scoped	2,681 indexed factor-base archives verify with NumPy installed
C13	broken	32 cited factor-base digests lack archive or declared debt
C14	broken	four complete scratch pilots lack spec and host audit inputs
C15	broken	13 interrupted sessions were counted as complete passes

Main limitations

The registry holds 2,753 experiment IDs and 3,726 run IDs in the auto researcher, 636 crypto session paths (617 ecbench/v1), and 335 cryptanalysis experiment directories. Only selected load-bearing joints received independent reading. The 84 checked targets cite only five exact experiment IDs and five full run IDs across their specs and 29 pinned source files; this is pointer coverage, not proof of those runs. The 617 ecbench/v1 directories have source-hashed integrity replays: 590 complete-status bound passes, 13 interrupted-status bound integrity passes for retained records only, ten interrupted and unbound, and four missing required source inputs. These counts do not replay timing and are not a theorem-to-run coverage fraction. The n=83 stack omits factor-base construction from both arms; a rank-stage planted control and cross-host replay are missing. The STARK public closure accepts an out-of-contract non-polynomial AIR, while the tested degree-one mutants reject. Any checker infrastructure timeout remains a coverage gap, not a mathematical counterexample. Pending PRs and public dashboard refresh are separate gates. After this frozen snapshot, the PR replayed all four amended BINSTD cells: 160 deterministic targets and four archived hit-count pairs matched under historical source commits and a separate exhaustive EC pair-sum oracle. Original per-target traces and host timing remain unavailable.